IRONFORGE MANUFACTURING LAB REPORT

Remote Login Success: IF-MAC-LEG01 into IF-WIN-HP01

Lab: IronForge Manufacturing Endpoint Connectivity Lab
Artifact Type: Remote login / SSH validation report
Source Endpoint: IF-MAC-LEG01
Destination Endpoint: IF-WIN-HP01
Protocol: SSH / Remote Login
Status: Successful remote login achieved, followed by continued path-resolution troubleshooting
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1. Executive Summary

This lab validated a successful remote login from the legacy macOS endpoint **IF-MAC-LEG01** into the Windows endpoint **IF-WIN-HP01**. The connection required accurate identification of the destination endpoint, correct user context, and use of the proper SSH login format.

The successful login represented a major troubleshooting milestone because previous attempts had produced errors such as hostname resolution failure, unreachable paths, and route-related failures. After the successful login, a later attempt using the same known-good path did not resolve as expected. This suggests that the issue may not be solely credential-based or command-format-based, but may involve network context, name resolution, VM networking behavior, endpoint state, or path interpretation between environments.

A key learning outcome was the value of using discovery and verification commands to retrieve endpoint identity, IP address, username, and network state instead of relying on memory or assumed names.

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2. Lab Objective

The objective of this lab was to prove whether **IF-MAC-LEG01** could remotely authenticate into **IF-WIN-HP01** using SSH / Remote Login over the local network.

The secondary objective was to document the troubleshooting process in a portfolio-ready format suitable for GitHub, showing:

- Endpoint identification
- SSH command structure
- Authentication context
- Successful login evidence
- Failure evidence after a later retest
- Lessons learned from path, hostname, and network troubleshooting
- Next steps for validating whether the VM layer is contributing to inconsistent results

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3. Environment Overview

Component	Role	Notes
IF-MAC-LEG01	Source endpoint	Legacy MacBook Pro used to initiate remote login
IF-WIN-HP01	Destination endpoint	Windows HP endpoint with OpenSSH Server enabled
SHELL-E	Main host / lab control station	Apple Silicon host used across IronForge lab work
Windows VM / Apex-style VM context	Variable network environment	Possible contributor to inconsistent path or route behavior
Local Router / LAN	Network layer	Used for DHCP, endpoint visibility, and local reachability checks

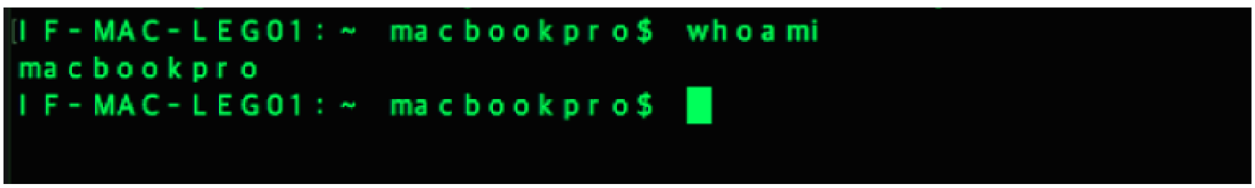

Figure 1 – Source endpoint identity validation: Terminal output confirming the active user account and host context on IF-MAC-LEG01 prior to initiating the remote login test.

A dedicated test account, sshlab, was used for remote login validation to avoid exposing or relying on a personal daily-use account.

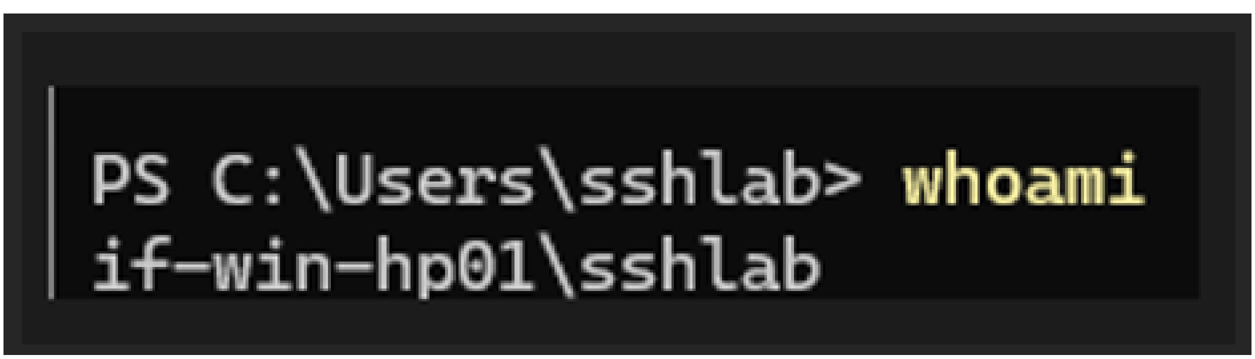

Figure 2 – Destination endpoint identity validation: Windows PowerShell output showing the whoami command executed on IF-WIN-HP01. The result confirms the active session is running under the dedicated sshlab test account before validating remote access behavior.

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4. Known-Good Connection Pattern

The expected SSH login structure is:

ssh <username_redacted>@ip_address

Example format:

ssh <WindowsUsername>@192.168.x.xxx

The successful connection depended on three separate identifiers being correct:

1. **Destination IP address**
   The current LAN address assigned to IF-WIN-HP01.
2. **Remote username**
   The Windows account name accepted by IF-WIN-HP01 for SSH login.
3. **Endpoint / host identity**
   The device name or host label used for recognition, documentation, and troubleshooting.

These are related but not interchangeable. A device name, a username, and an IP address each serve different purposes.

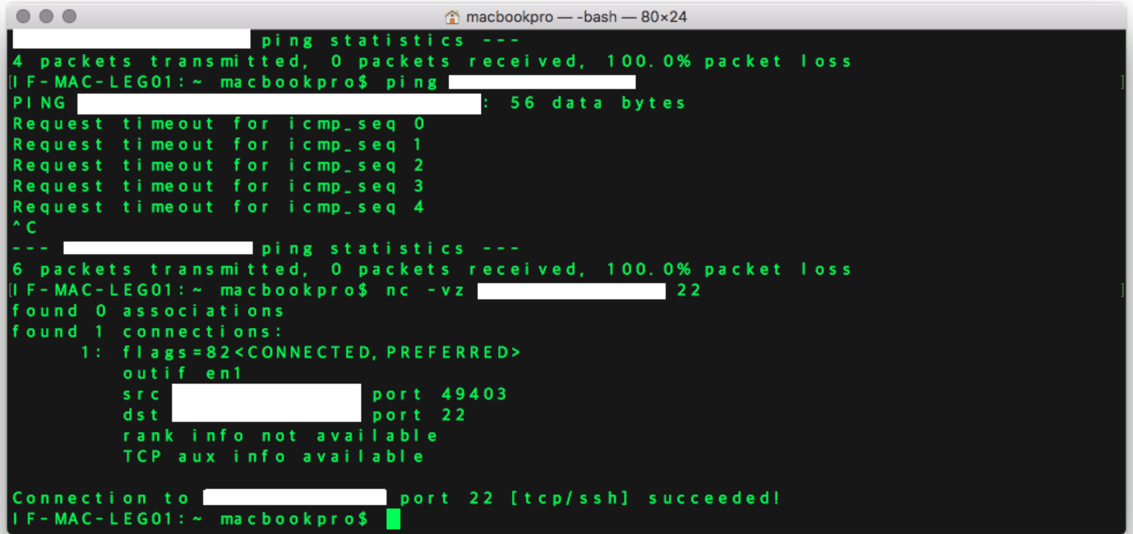

Figure 3 – ICMP failure with TCP/SSH reachability confirmed: Ping from IF-MAC-LEG01 to IF-WIN-HP01 returned 100% packet loss, while a Netcat test to TCP port 22 succeeded. This indicates ICMP echo replies were unavailable or filtered, but the SSH service remained reachable over TCP.

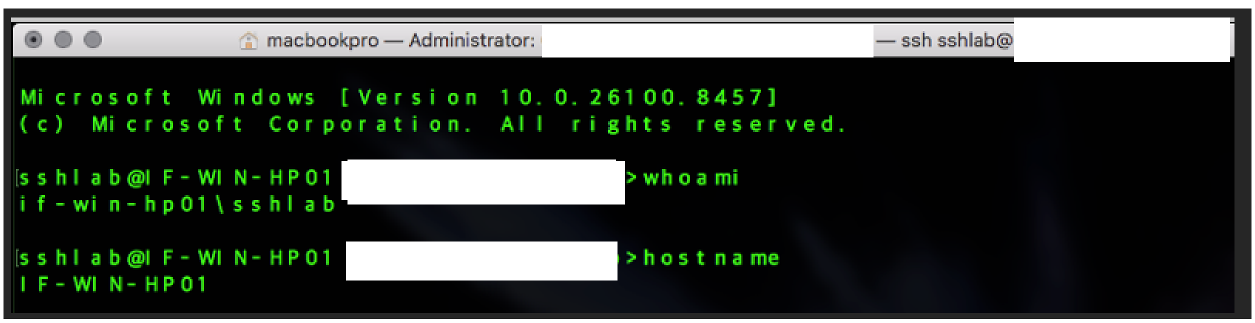

Figure 4 – Successful SSH remote session verification: After authenticating from IF-MAC-LEG01 into IF-WIN-HP01, the remote shell confirmed the active Windows user context and destination hostname using whoami and hostname.

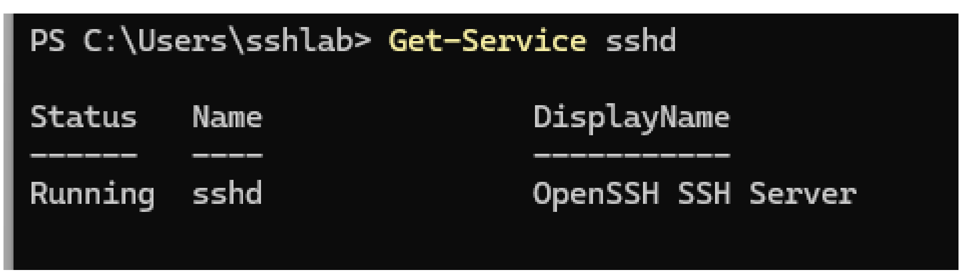

Figure 5 – OpenSSH service validation: PowerShell output on IF-WIN-HP01 confirming the sshd service was running as the OpenSSH SSH Server during remote access testing.

```
PS C:\Users\sshlab> netstat -ano | findstr :22
  TCP    0.0.0.0:22             0.0.0.0:0              LISTENING       4440
  TCP    [::]:22                [::]:0                 LISTENING       4440
PS C:\Users\sshlab> |
```

Figure 6 – TCP/22 listener validation: The netstat -ano | findstr :22 output confirms IF-WIN-HP01 was actively listening on TCP/22, supporting the successful SSH connection observed during remote login testing.

```
    Directory: C:\Users\sshlab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-r---         5/17/2026  10:04 AM                Contacts
d-r---         5/17/2026  10:04 AM                Desktop
d-r---         5/17/2026  10:04 AM                Documents
d-r---         5/17/2026  10:04 AM                Downloads
d-r---         5/17/2026  10:04 AM                Favorites
d-r---         5/17/2026  10:04 AM                Links
d-r---         5/17/2026  10:04 AM                Music
d-r---         5/17/2026  10:07 AM                OneDrive
d-r---         5/17/2026  10:37 AM                Pictures
d-----         5/17/2026   6:41 PM                Remote_login
d-r---         5/17/2026  10:04 AM                Saved Games
d-r---         5/17/2026  10:04 AM                Searches
d-r---         5/17/2026  10:04 AM                Videos
-a----         5/15/2026   9:38 PM             35 ssh_success.txt


PS C:\Users\sshlab> cat ssh_success.txt
SSH SUCCESS from LEG01 into HP01
PS C:\Users\sshlab>
```

Figure 7 – SSH success artifact validation: PowerShell output confirming that ssh_success.txt was present in the sshlab user directory and contained a note documenting successful SSH access from LEG01 into HP01.

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5. Why There Are Multiple "Names" Involved

During troubleshooting, it became clear that several identifiers can appear to describe the same machine, but each one belongs to a different layer.

Identifier	Example	Purpose
Computer / host name	IF-WIN-HP01	Identifies the device on the network or OS

Identifier	Example	Purpose
Username	******* / admin / local account name	Identifies the account being authenticated into
IP address	192.168.x.xxx	Identifies the network location of the endpoint
Terminal prompt path	<username_redacted> or shell prompt	Shows where the session is currently operating after login

The remote login command does **not** use the computer name by itself unless DNS / local name resolution is working correctly. The most reliable format during this lab was:

ssh username@ip_address

This bypasses some hostname resolution issues by targeting the endpoint's IP address directly.

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6. Troubleshooting Summary

6.1 Issues Encountered

Several obstacles appeared before the successful login:

- Hostname could not be resolved
- Incorrect or uncertain endpoint names
- Confusion between username, device name, and terminal path
- Route / reachability errors
- Password versus PIN confusion on Windows
- Need to confirm OpenSSH Server status on IF-WIN-HP01
- Need to verify port 22 availability
- Possible inconsistency caused by VM networking mode or route behavior

6.2 What Worked

The successful breakthrough came from verifying facts directly instead of relying on remembered values:

- Confirming the correct IP address
- Confirming the correct remote username
- Using the correct SSH command format
- Distinguishing Windows sign-in PIN from account password
- Testing the connection from IF-MAC-LEG01 into IF-WIN-HP01

- Reaching an authenticated shell session on the Windows endpoint

6.3 What Did Not Fully Resolve

After the successful login, repeating the same path later did not produce the same result. The system still failed to find the path during a later attempt.

This suggests that the environment may have at least one variable still changing between attempts, such as:

- IP lease change
- Endpoint sleep / power state
- Network profile change
- Wi-Fi band / SSID behavior
- Hostname resolution inconsistency
- Firewall state
- SSH service state
- VM network mode changing route assumptions
- Attempting the connection from a different network context than the successful test

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7. Assessment: Expected Troubleshooting vs. Knowledge Gap

Expected Troubleshooting

A significant portion of the difficulty was normal and expected for a cross-platform remote login lab. This is especially true because the lab involved:

- macOS to Windows SSH
- Legacy endpoint behavior
- Windows OpenSSH configuration
- Local network routing
- Firewall considerations
- Account/password differences
- Dynamic addressing
- VM networking variables

These are realistic conditions. Endpoint connectivity work often requires confirming each layer before assuming the failure is credential-related.

Knowledge Gap / Learning Curve

Some of the delay came from still building fluency with the differences between:

- Device name vs. username
- Local terminal prompt vs. remote shell
- PIN vs. password
- Hostname vs. IP address
- Network reachability vs. authentication failure
- VM network path vs. physical endpoint LAN path

This was not a failure of the lab. It was the exact learning value of the lab. The exercise converted abstract networking and identity concepts into hands-on troubleshooting experience.

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8. VM Networking Hypothesis

The later failure to find the same path raises a valid hypothesis: the VM environment may be contributing to inconsistent network behavior.

VMs can introduce additional complexity because they may use different network modes, including:

- Shared / NAT networking
- Host-only networking
- Bridged networking
- Isolated lab networks

If the successful test occurred from a physical endpoint on the LAN, but the later test involved a VM or a different route context, the traffic path may not be equivalent.

Working Hypothesis

The successful IF-MAC-LEG01 → IF-WIN-HP01 connection proves that the Windows endpoint can accept remote login under the correct conditions. The later failure suggests that the unresolved issue may be environmental rather than purely command-related.

The VM should be treated as a separate network zone until proven otherwise.

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9. Evidence Collected for GitHub

The following evidence categories were documented to support the remote login validation report. The published GitHub repository includes the full redacted report, selected supporting screenshots, and a troubleshooting notes artifact. Some command outputs are preserved visually within the report figures rather than as separate raw text files.

Evidence Category	Purpose
Source endpoint identity validation	Shows IF-MAC-LEG01 user, host, and local terminal context prior to initiating the remote login test.
Destination endpoint identity validation	Shows IF-WIN-HP01 destination endpoint context and confirms the dedicated sshlab test account used for validation.
ICMP reachability failure	Documents the ICMP reachability test from IF-MAC-LEG01 to IF-WIN-HP01 showing 100% packet loss.
TCP/22 validation	Shows TCP/22 service validation from IF-MAC-LEG01, confirming SSH-specific connectivity was available even though ICMP ping failed.
SSH session verification	Captures the successful SSH remote login from IF-MAC-LEG01 into IF-WIN-HP01 and verifies the remote user context and hostname.
OpenSSH service status	Confirms the Windows OpenSSH Server service, sshd, was running on IF-WIN-HP01 during remote access testing.
TCP/22 listener validation	Shows IF-WIN-HP01 actively listening on TCP port 22 for SSH connections using netstat.
SSH success artifact validation	Confirms the ssh_success.txt file was present in the sshlab user directory and contained a note documenting successful SSH access from LEG01 into HP01.
Troubleshooting notes	Summarizes the test path, key findings, troubleshooting logic, and final conclusion for GitHub review.

9.1 Published GitHub Artifacts

Published Artifact	Purpose
README.md	Provides the project overview, key finding, skills demonstrated, repository contents, and direct links.
report/Redacted_Remote_Login_original_Final.pdf	Contains the complete redacted report and full figure sequence.

Published Artifact	Purpose
screenshots/	Contains selected supporting screenshots from the validation process.
evidence/09_remote_login_troubleshooting_notes.md	Provides a written summary of scope, key findings, troubleshooting logic, and conclusion.

9.2 Screenshot and Evidence Mapping

Figure / Evidence Area	Published Location	Purpose
Figure 1 – Source endpoint identity validation	report/Redacted_Remote_Login_original_Final.pdf	Documents IF-MAC-LEG01 user, host, and local terminal context prior to initiating the remote login test.
Figure 2 – Destination endpoint identity validation	report/Redacted_Remote_Login_original_Final.pdf	Documents IF-WIN-HP01 destination endpoint context and confirms the dedicated sshlab test account used for validation.
Figure 3 – ICMP failure with TCP/22 reachability confirmed	report/Redacted_Remote_Login_original_Final.pdf and screenshots/	Shows ICMP ping failure while TCP/22 validation confirmed SSH-specific connectivity was available.
Figure 4 – Successful SSH remote session verification	report/Redacted_Remote_Login_original_Final.pdf and screenshots/	Confirms successful SSH authentication from IF-MAC-LEG01 into IF-WIN-HP01 and verifies remote user context and hostname.
Figure 5 – OpenSSH service validation	report/Redacted_Remote_Login_original_Final.pdf and screenshots/	Confirms the Windows OpenSSH Server service, sshd, was running during remote access testing.
Figure 6 – TCP/22 listener validation	report/Redacted_Remote_Login_original_Final.pdf	Shows IF-WIN-HP01 actively listening on TCP port 22 for SSH connections.
Figure 7 – SSH success artifact validation	report/Redacted_Remote_Login_original_Final.pdf	Confirms the ssh_success.txt artifact was

Figure / Evidence Area	Published Location	Purpose
		present and documented successful SSH access from LEG01 into HP01.
Troubleshooting notes	evidence/09_remote_login_troubleshooting_notes.md	Provides a written summary of scope, key findings, troubleshooting logic, and final conclusion.

Together, these artifacts show the full validation path: endpoint identity was confirmed, ICMP failure was documented, TCP/22 reachability was verified, SSH authentication succeeded, the destination service state was validated, and a success artifact was preserved for review.

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10. Conclusion

This lab confirmed successful SSH-based remote login from IF-MAC-LEG01 into IF-WIN-HP01. The most important technical finding was that ICMP ping returned 100% packet loss, but SSH authentication over TCP/22 still succeeded. This demonstrated that ping failure did not equal complete endpoint failure and that protocol-specific validation was required.

The lab also reinforced the importance of separating endpoint identity, username, IP address, service state, and terminal path. Each layer had to be verified independently before the remote login path could be understood with confidence.

A later inconsistent path result remains under review as part of the VM networking hypothesis. Until the VM network mode and route behavior are validated, the VM should be treated as a separate network context from the physical LAN endpoints.

Overall, this exercise strengthened practical troubleshooting skills across macOS, Windows, SSH, endpoint identity, and local network validation. The collected evidence supports the successful remote login while preserving unresolved behavior for continued investigation.

11. Portfolio Summary

This project demonstrates practical endpoint troubleshooting by validating SSH-based remote login from a legacy macOS endpoint into a Windows endpoint. The work included endpoint identification, service-specific connectivity testing, authentication validation, evidence collection, and analysis of inconsistent network-path behavior. The result shows the ability to distinguish protocol-specific reachability from general ICMP failure and preserve audit-ready technical evidence.